

CLASS XII · LECTURE 11

Visual Basic Programs for the Examination

Option II · A practical lecture · Write each program from a blank page

By the end of this lecture you can

- Write the even-number loop and the ten-number average from memory.
- Write Select Case and a Function that returns a value.
- Name the lines a marker expects in each.

The set to be able to reproduce

Four pieces cover most programming questions in Option II. Database questions in the same paper are prepared in Lectures 12 to 14.

Even numbers from 0 to 50. Program 9.2. The mark is Step 2. A loop For i = 0 To 50 without Step prints the odd numbers as well and does not answer the question. Starting at 1 with Step 2 prints the odds.

Sum and average of ten numbers. Program 9.3. total must start at 0. The average is total / 10, not total / i after the loop, because after Next the counter has moved on. Val around InputBox is part of the marks.

Select Case. Program 9.1. Include Case Else and End Select. Do not import break from C.

A function. Program 10.1. Assign the result to the function's name. The Click procedure only collects input and displays what the function returns.

EXAM TIP A program layout that is easy to mark

First line: Private Sub Command1_Click() or Function Then Dim. Then input. Then the loop or the Select. Then the output. Then End Sub or End Function. Indent the inside of the loop by two spaces. The marker is reading fifty scripts.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Short answers

S1. Write a VB program to print all even numbers from 0 to 50.

S2. Write a VB program for the sum and average of ten inputted numbers.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

S1. In Command1_Click, Dim i As Integer, then For i = 0 To 50 Step 2, Print i, Next i, End Sub. Step 2 is required. The default step of 1 would also print the odd numbers.

S2. Dim i As Integer, n As Double, total As Double. Set total = 0. For i = 1 To 10, add Val(InputBox(...)) to total. After Next, Print the total and total / 10.